Work 1 — Secure REST API Development with CI/CD and Real CVE Vulnerability Analysis

Work no. : 1

Title : Secure REST API Development with CI/CD and Real CVE Vulnerability Analysis

Variant : —

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Repository : `https://github.com/faris-faris-faris/work1-secure-notes-api`

Pipeline : `https://github.com/faris-faris-faris/work1-secure-notes-api/actions/runs/40103379944`

1. Purpose

1. Build a REST API from scratch with the protections the OWASP Top 10 asks for — parameterised SQL, output encoding, hashed passwords, verified JWTs and ownership checks — and prove each protection with a test rather than a claim.
2. Put SAST and SCA into CI/CD so every push and pull request is checked automatically, and show the tools' own output.
3. Reproduce a real CVE in a controlled Vulhub environment, explain why the defect exists, patch it, and demonstrate that the attack no longer works.
4. Present the result in an engineering form: reproducible commands, comparable before/after evidence, and a fix verified by behaviour rather than by a status code.

2. Environment

Item	Version / notes
OS	Windows 11 (26300); CI runs on `ubuntu-latest`
Runtime	Node.js 24.16 (CI matrix: Node 22 and 24)
Framework	Express 5.3
Database	SQLite via the built-in `node:sqlite` module
Validation	Zod 4.6
Auth	scrypt ($N=2^{15}$) + HS256 JWT (`jsonwebtoken` 9.0) + rotating refresh tokens
Tests	`node:test` + `supertest`, 52 cases
SAST	ESLint 9 + `eslint-plugin-security`
SCA	`npm audit` (CI) / OSV.dev (`scripts/sca-osv.mjs`)
Containers	Docker Desktop 29.8, Docker Compose v2
CVE range	Vulhub `spring/CVE-2022-22965` (Spring 5.3.17 → 5.3.18)

Why Node.js and not Flask. The previous piece of coursework in this course was a Python/Flask service; using Node.js + Express here produces a genuinely different implementation instead of a re-typed one. It also exercises a different set of failure modes — prototype/JSON handling, `ORDER BY` cannot be parameterised the same way, and the password-hashing library is different (scrypt from the standard library) — while still covering every requirement of the assignment.

3. Part 1 — the secure REST API

3.1 Stack and initialisation

```
mkdir work1-secure-notes-api && cd work1-secure-notes-api
npm init -y
npm install express@5 jsonwebtoken zod
npm install -D eslint eslint-plugin-security supertest
git init && git add . && git commit -m "feat: secure notes API skeleton"
git remote add origin https://github.com/faris-faris-faris/work1-secure-notes-api.git
git push -u origin main
```

```
src/
├── app.js              application factory: headers, JSON limit, error handler
├── server.js           process entry point, graceful shutdown
├── config.js           environment configuration + production guards
├── db.js               schema and every SQL statement
├── security.js         scrypt, JWT, output encoding, rate limiter
├── validation.js       Zod schemas for every request
├── errors.js           one client-safe error type
├── middleware/auth.js  JWT verification, ownership helper
└── routes/             auth.js, notes.js
tests/                  52 tests
cve/                    vulnerable environment, exploit, patch, verification
.github/workflows/ci.yml
```

3.2 API design

#	Method	Path	Auth	Purpose
1	POST	`/auth/login`	—	Accepts `login` / `password`, returns an access token and a refresh token
2	GET	`/api/data`	Bearer	Note feed (`limit` , `offset` , `q` , `category` , `sort`)
3	POST	`/api/notes`	Bearer	**Own method**: create a note
+	GET	`/api/notes/:id`	Bearer	Read one note
+	PATCH	`/api/notes/:id`	Bearer	Edit (author or admin only)
+	DELETE	`/api/notes/:id`	Bearer	Delete (author or admin only)
+	POST	`/auth/refresh`	—	Rotate the refresh token
+	POST	`/auth/logout`	—	Revoke the refresh token
+	GET	`/api/me`	Bearer	Identity + token metadata
+	GET	`/health`	—	Liveness probe

```
curl -s -X POST http://127.0.0.1:3000/auth/login \
     -H 'Content-Type: application/json' \
     -d '{"login":"alice","password":"Passw0rd!Alice"}'
```

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "expires_in": 900,
  "refresh_token": "8Kx…",
  "user": {"id": 1, "login": "alice", "display_name": "Alice (course admin)", "role": "admin"}
}
```

Without a token:

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Bearer error="missing_token"
Content-Type: application/json; charset=utf-8

{"error":{"code":"missing_token","message":"Missing or malformed Authorization header (expected
\"Bearer <token>\")."}}
```

Measured walkthrough — `scripts/run-demo.ps1`, 16 checks, all `PASS`:

Step	Request	Result
1	`GET /health`	200
2	`POST /auth/login` (wrong password)	401
3	`POST /auth/login` (alice)	200, JWT + refresh token, `expires_in=900`
4	`GET /api/data` without a token	**401** **`missing_token`**
5	`GET /api/data` with a forged token	**401** **`invalid_token`**
6	`GET /api/data` with the real token	200, `total=3`
7	`POST /api/notes`	**201**, `Location: /api/notes/4`
8	`PATCH` that note as `bob`	**403 forbidden**
9	login with `alice' OR '1'='1`	**401**
10	`GET /api/data?q=' OR 1=1 --`	200, `total=0` (literal text)
11	store `<script>alert("XSS")</script>`	201, returned as `<script>…`
12	refresh, then replay the old token	200, then **401** **`token_reuse`**
13	`PATCH` + `DELETE` own note	200, 204
14	six failed logins	**429**, `Retry-After: 300`

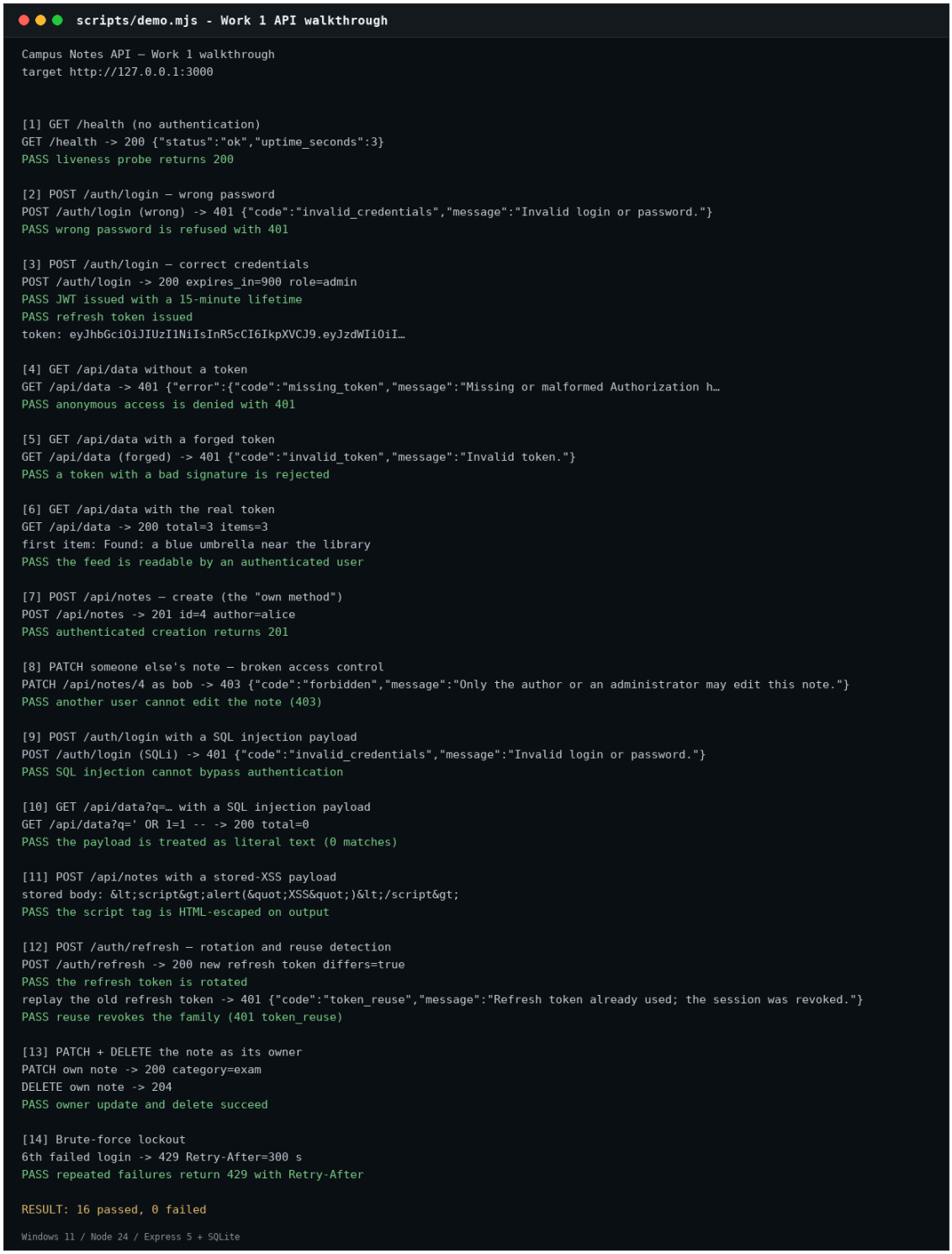


Figure 1 — the complete walkthrough. Every line is the response of a real request against a real SQLite database.

3.3 SQL injection

Rule 1: every value is a bound parameter. `src/db.js` is the only module that contains SQL, which makes the rule reviewable in one file:

```
const user = db.prepare('SELECT * FROM users WHERE login = ? COLLATE NOCASE').get(login);
```

Rule 2: `LIKE` needs its own escaping. A bound parameter stops SQL syntax injection, but `%` and `_` inside the *value* are still wildcards. A user searching for `100%` should not receive the whole table, so the term is escaped and the statement declares the escape character:

```
function likePattern(term) {
  return `%${term.replace(/[\\%_]/g, (char) => `\\${char}`)}%`;
}
// ...
`(n.title LIKE ? ESCAPE '\\' OR n.body LIKE ? ESCAPE '\\')`
```

Rule 3: identifiers cannot be parameterised, so they are allow-listed. Sort order comes from the client but never reaches SQL text:

```
const SORT_CLAUSES = Object.freeze({
  newest: 'n.created_at DESC, n.id DESC',
  oldest: 'n.created_at ASC, n.id ASC',
  title:  'n.title COLLATE NOCASE ASC, n.id ASC',
});
const orderBy = SORT_CLAUSES[sort] || SORT_CLAUSES.newest;
```

Validation rejects anything outside the enum before the repository is called, so `?sort=id; DROP TABLE notes` is a `400`, not a query.

Evidence. `tests/notes.test.js` sends `' OR 1=1 --`, `' UNION SELECT * FROM users --` and `'; DROP TABLE notes; --` to the search endpoint: every one returns `total=0`, and a following assertion checks that the `notes` table still has its rows. The login endpoint refuses the tautology payload with `401`. A separate test searches for `%` and gets exactly the one note that contains a literal percent sign.

3.4 Cross-site scripting

The database stores text verbatim and the API encodes it **on output** — the rule from the OWASP XSS Prevention Cheat Sheet. Encoding on input would corrupt the data (and double-encode on the next edit); encoding on output keeps the text usable by a JSON client while making it inert for an HTML one.

```
const HTML_ESCAPES = { '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;', '`':
'&#96;' };
export const escapeHtml = (value) => String(value).replace(/[&<>"'`]/g, (c) => HTML_ESCAPES[c]);
```

Every user-controlled field of a note (`title`, `body`, author login and display name) passes through `escapeHtml` in `presentNote()`; the test stores `<script>`, `<img onerror>`, `<svg onload>`, `javascript:` and `<iframe>` payloads and asserts that none of the raw forms appears in any response, while the database still contains the original text (checked directly against SQLite).

Defence in depth:

- `Content-Type: application/json` and `X-Content-Type-Options: nosniff`;
- `Content-Security-Policy: default-src 'none'; frame-ancestors 'none'` — even a document that rendered a payload could not execute script;
- validation rejects control characters and caps each field (`title` 120, `body` 4000), which also blocks log-injection attempts;
- unknown JSON fields are rejected (`.strict()`), so a client cannot smuggle `author_id` or `role` into a create request.

3.5 Authentication

Requirement	Implementation
Never store plaintext	scrypt, N=2^{15}, r=8, p=1, 16-byte salt; parameters stored with the hash so the cost can be raised later
Constant-time check	`crypto.timingSafeEqual` on the derived keys
Issue a token	HS256 JWT: `sub`, `iss`, `aud`, `iat`, `nbf`, `exp`, `jti`; 15-minute TTL
Verify on every protected endpoint	`makeAuthMiddleware` — signature, algorithm, issuer, audience, expiry, `sub`
Algorithm pinning	`algorithms: ['HS256']`, which defeats `alg: none` and RS→HS confusion
Revocation	the user is re-read from the database on every request; a deleted account's token dies immediately
No enumeration	the same 401 body for unknown user and wrong password, plus a dummy scrypt run to flatten timing
Brute force	5 failures per (IP, login) per 5 minutes → `429` with `Retry-After`
Refresh	256-bit opaque token, SHA-256 at rest, rotated on every use; replay revokes the family

The refresh design deserves a note, because it is where most "JWT + refresh" implementations go wrong. A refresh token is not a second JWT: it is an opaque random string whose hash is stored in the database. Each use rotates it — the old row is marked `rotated_at`, a new row is inserted in the same `family_id`. If a rotated token is presented again, either the client replayed a request or the token was stolen; the server revokes the whole family, so the attacker's copy and the victim's session die together:

```
if (row.rotated_at) {
  repo.revokeRefreshFamily(row.family_id);
  throw unauthorized('Refresh token already used; the session was revoked.', 'token_reuse');
}
```

Evidence: `tests/auth.test.js` covers expired tokens, attacker-signed tokens, `alg: none`, a foreign audience, deleted users, lockout and the hash format; `tests/refresh.test.js` covers rotation, reuse detection and logout.

3.6 Access control (A01)

Every write is checked against the loaded row, not against the token alone:

```
const existing = repo.getNote(id);
if (!existing) throw notFound('Note not found.');
if (!canModifyNote(req.user, existing)) {
  throw forbidden('Only the author or an administrator may edit this note.');
}
```

`canModifyNote` is `note.author_id === user.id || user.role === 'admin'`, and it lives in one place so a new route cannot forget it. The test suite creates a third user and asserts that editing another user's note is `403`

while an admin can moderate it. A non-numeric id (`1 OR 1=1`) is a `404` , not a database error.

3.7 Tests

```
npm test                  # 52 tests
npm run coverage          # 96.7 % lines, 86.0 % branches, 95.2 % functions
```

```
tests 52 | suites 6 | pass 52 | fail 0
all files | 96.73 % lines | 86.00 % branches | 95.18 % functions
```

Each test file builds its own application with an in-memory database, so tests share no state and can run in any order. They exercise the real Express stack and the real SQL engine; nothing is mocked except the logger.



Figure 2 — the suite, run by `node --test`.

3.8 CI/CD with SAST and SCA

`.github/workflows/ci.yml` runs on every push and pull request:

Job	Tool	What it proves
`test`	`node --test`, Node 22 + 24 matrix, coverage	functional and security regressions pass on both LTS lines
`lint`	ESLint `--max-warnings=0`	unused code, obvious bug patterns
`sast`	ESLint + `eslint-plugin-security`	no `eval`, dynamic `require`, unsafe regex, `child_process` misuse, …
`sca`	`npm audit --audit-level=high`	no known vulnerable dependency in the lockfile

Both scanners write their output to the job summary and upload it as an artifact, so a reviewer can see the tool output rather than a green checkmark.

The same four jobs can be reproduced locally with `scripts/ci-local.ps1` (`npm ci` is skipped because the tree is already installed):

```
Job  Exit Status
test    0 PASS
lint    0 PASS
sast    0 PASS
sca     0 PASS
RESULT: all 4 jobs passed
```

ci.yml - all four jobs, run locally

```
=== job: test (node --test) ===
▶ authentication
  ✔ issues a JWT and a refresh token for valid credentials (97.1521ms)
  ✔ accepts the login case-insensitively (16.2595ms)
  ✔ rejects a wrong password with 401 (16.3281ms)
  ✔ answers unknown user and wrong password identically (no enumeration) (34.8082ms)
  ✔ rejects a missing field with a validation error (10.1338ms)
  ✔ rejects unknown extra fields (mass assignment guard) (8.1798ms)
  ✔ denies a protected endpoint without a token (8.233ms)
  ✔ denies a malformed Authorization header (8.0293ms)
  ✔ denies a token signed with another key (11.3559ms)
  ✔ denies an unsigned (alg: none) token (8.8766ms)
  ✔ denies an expired token (8.0101ms)
  ✔ denies a token issued for another audience (8.4192ms)
  ✔ stops working after the account is deleted (26.1546ms)
  ✔ locks out after the configured number of failed attempts (56.4188ms)
  ✔ never stores the password in plaintext (0.6672ms)
✔ authentication (350.7636ms)
▶ password hashing primitives
  ✔ produces a unique salt per hash (19.5476ms)
  ✔ rejects a wrong password and malformed hashes (9.6238ms)
  ✔ escapes the HTML-significant characters (0.2414ms)
✔ password hashing primitives (29.7394ms)
▶ notes API
  ✔ returns the feed to an authenticated user (14.1679ms)
  ✔ paginates deterministically (16.5475ms)
  ✔ sorts only by whitelisted columns (35.4656ms)
  ✔ treats a SQL injection payload in the search term as literal text (22.2035ms)
  ✔ escapes LIKE wildcards so "%" is a literal character (20.5862ms)
  ✔ filters by category (7.7181ms)
  ✔ rejects unknown query parameters (7.2575ms)
  ✔ creates a note and returns 201 with a Location header (6.895ms)
  ✔ rejects invalid and unknown fields on create (25.8482ms)
  ✔ rejects control characters in text fields (7.5563ms)
  ✔ lets the author edit a note (7.1357ms)
  ✔ forbids editing someone else's note (IDOR) (7.2359ms)
  ✔ lets an administrator moderate any note (6.6835ms)
  ✔ returns 404 for a note that does not exist or a non-numeric id (17.2135ms)
  ✔ deletes a note for its author and then 404s (21.7135ms)
  ✔ escapes stored XSS payloads on output while storing the raw text (27.0439ms)
  ✔ reports the identity carried by the token (5.5972ms)
  ✔ rejects an oversized body with 413 (9.3272ms)
  ✔ rejects malformed JSON with 400 (6.2314ms)
✔ notes API (446.7416ms)
▶ refresh-token sessions
  ✔ rotates the refresh token on every use (111.4361ms)
  ✔ revokes the whole family when a rotated token is replayed (40.6981ms)
  ✔ rejects an unknown or malformed refresh token (16.604ms)
  ✔ rejects an expired refresh token (21.7118ms)
  ✔ revokes the session on logout (26.0861ms)
✔ refresh-token sessions (239.9597ms)
▶ HTTP hardening
  ✔ sets the security headers on success responses (12.6799ms)
  ✔ sets the same headers on errors and 404s (19.4645ms)
  ✔ does not advertise the framework (8.6858ms)
  ✔ does not require authentication for the liveness probe (7.7802ms)
  ✔ returns a JSON 404 for an unknown endpoint (7.5171ms)
  ✔ does not allow an unsupported method on an existing path (7.3567ms)
✔ HTTP hardening (189.8529ms)
▶ configuration guards
  ✔ refuses to start in production without a strong JWT secret (1.8233ms)
  ✔ refuses demo users in production (0.4961ms)
  ✔ generates a fresh random secret in development (0.5181ms)
  ✔ rejects an invalid numeric setting (0.4131ms)
✔ configuration guards (3.9998ms)
ℹ tests 52
ℹ suites 6
ℹ pass 52
ℹ fail 0
ℹ cancelled 0
ℹ skipped 0
ℹ todo 0
ℹ duration_ms 2167.6618

=== job: lint (eslint --max-warnings=0) ===

=== job: sast (eslint-security) ===
SAST report — ESLint with eslint-plugin-security
files analysed      : 19
errors              : 0
warnings            : 0
distinct rules hit  : 0

findings by rule: none

RESULT: no errors — the SAST gate passes.

=== job: sca (OSV.dev — CI runs npm audit) ===
SCA report — OSV.dev (aggregates GHSA / CVE / ecosystem advisories)
runtime dependencies : 83
dev dependencies     : 107 (not shipped, still scanned for completeness)

No known vulnerabilities found in 190 locked packages.
RESULT: SCA gate passes.

=== pipeline summary ===


Job  Exit Status

---  ---- ------

test    0 PASS

lint    0 PASS

sast    0 PASS

sca     0 PASS




RESULT: all 4 jobs passed
```

scripts/ci-local.ps1

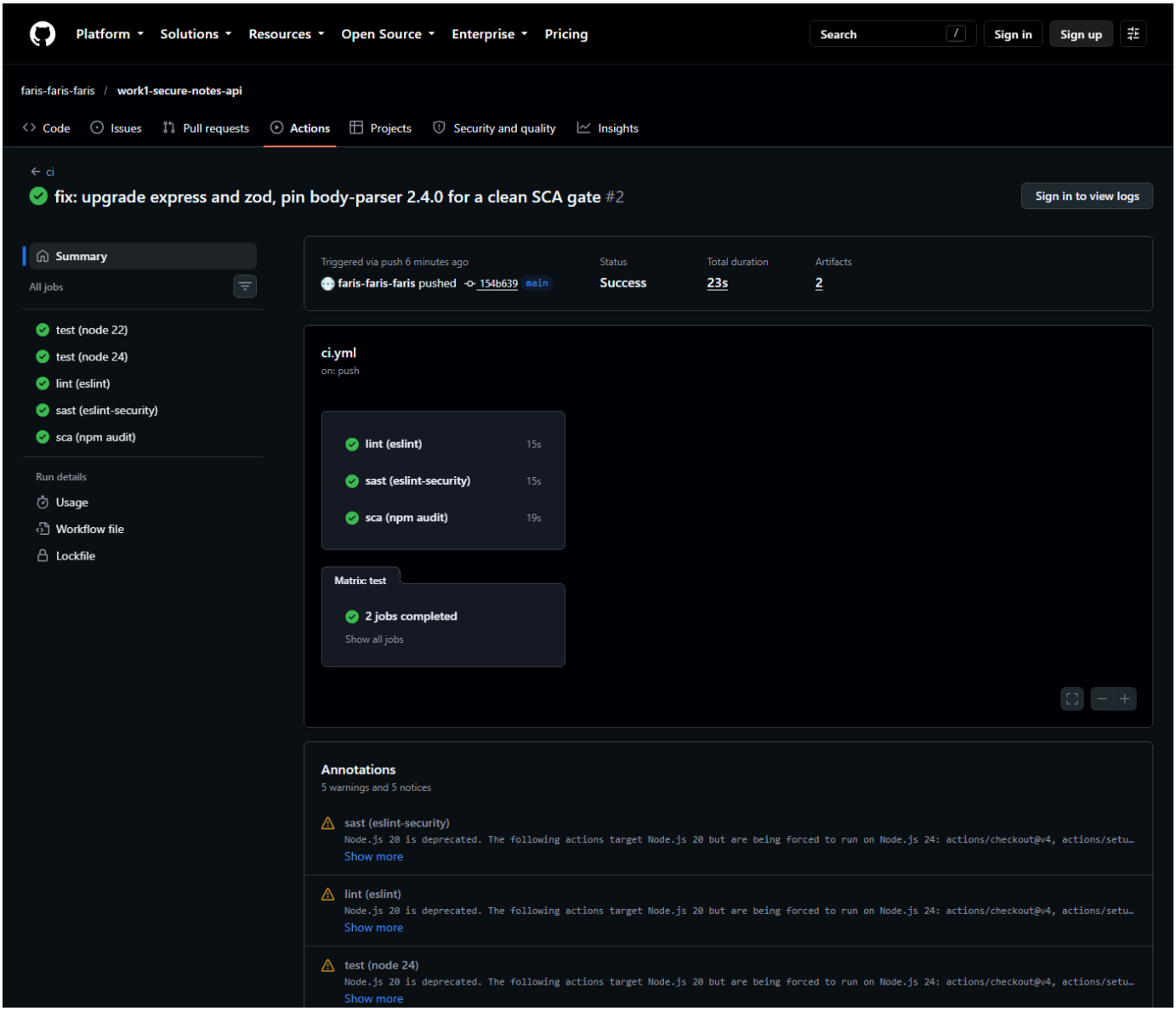


Figure 3 — the four `ci.yml` jobs: reproduced locally (top) and green on GitHub Actions, run 40103379944 on commit `154b639` (bottom).

SAST - ESLint + eslint-plugin-security

```
SAST report — ESLint with eslint-plugin-security
files analysed      : 19
errors              : 0
warnings            : 0
distinct rules hit  : 0

findings by rule: none

RESULT: no errors — the SAST gate passes.

npx eslint . --format json
```

Figure 4 — SAST report: 19 files, 0 errors, 0 warnings.

SCA - OSV.dev dependency scan

```
SCA report — OSV.dev (aggregates GHSA / CVE / ecosystem advisories)
runtime dependencies : 80
dev dependencies     : 107 (not shipped, still scanned for completeness)

No known vulnerabilities found in 187 locked packages.
RESULT: SCA gate passes.

node scripts/sca-osv.mjs
```

Figure 5 — SCA report: 187 locked packages, no known vulnerabilities.

> **On the pipeline link.** The top half of Figure 3 is a local run of the exact commands in `ci.yml`; the bottom half is the real GitHub Actions run **40103379944** on commit `154b639`, where `test` on Node 22 and 24, `lint`, `sast` and `sca` are all green. The raw run data is recorded in `docs/evidence/github-actions-run.txt`; the consolidated pipeline link is on the title page and in §7.

3.9 Known limitations

- The rate limiter and the session store are in-process: correct for one instance, but a multi-instance deployment needs Redis (or a shared table) so limits and revocations are global.
- Refresh tokens are stored hashed but not bound to a device; a stolen token that is used *before* rotation cannot be distinguished from the real client.
- The API has no CORS layer because no browser front end is part of this work. A real deployment needs an explicit allow-list.
- SQLite serialises writes; that is ideal for a lab and wrong for a write-heavy production service.

---

4. Part 2 — CVE-2022-22965 (Spring4Shell)

4.1 Selection

Property	Value
CVE	**CVE-2022-22965**
Component	Spring Framework MVC (data binding), WAR on Tomcat, JDK 9+
Class	CWE-94 — Improper Control of Generation of Code ("Code Injection")
CVSS 3.1	**9.8 CRITICAL** — `AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H`
Affected	Spring 5.3.0–5.3.17, 5.2.0–5.2.19
Fixed in	Spring 5.3.18 / 5.2.20 (2022-03-31)
Vulhub	`vulhub/spring/CVE-2022-22965`

The CVE was chosen because the root cause is a *framework design* problem rather than a missing check in one application, because the exploit is a single HTTP request whose effect is unambiguous, and because the fix is a version upgrade — exactly the remediation path the lab asks the student to practise.

4.2 Vulnerable environment

```
cd cve
docker compose up -d          # vulhub/spring-webmvc:5.3.17
```

Inside the container: Spring MVC 5.3.17 packaged as `ROOT.war`, Tomcat 8.5.77, OpenJDK 11. The WAR deployment is not incidental — see §4.4.

```
curl -s 'http://127.0.0.1:8080/?name=Bob&age=25'
# <h2>Hello, my name is Bob, I am 25 years old.</h2>
```

4.3 Reproducing the attack

```
powershell -ExecutionPolicy Bypass -File .\exploit.ps1
```

```
1. Baseline: the application works normally
   <h2>Hello, my name is Bob, I am 25 years old.</h2>

2. Exploit: rewrite the Tomcat access-log valve into a JSP webshell
   HTTP 200

3. Trigger execution through the webshell: /tomcatwar.jsp?pwd=j&cmd=id
   uid=0(root) gid=0(root) groups=0(root)
   RCE CONFIRMED — the command output above was produced by the target process
   --- whoami ---
   root
   --- hostname ---
   fca45319547f
```

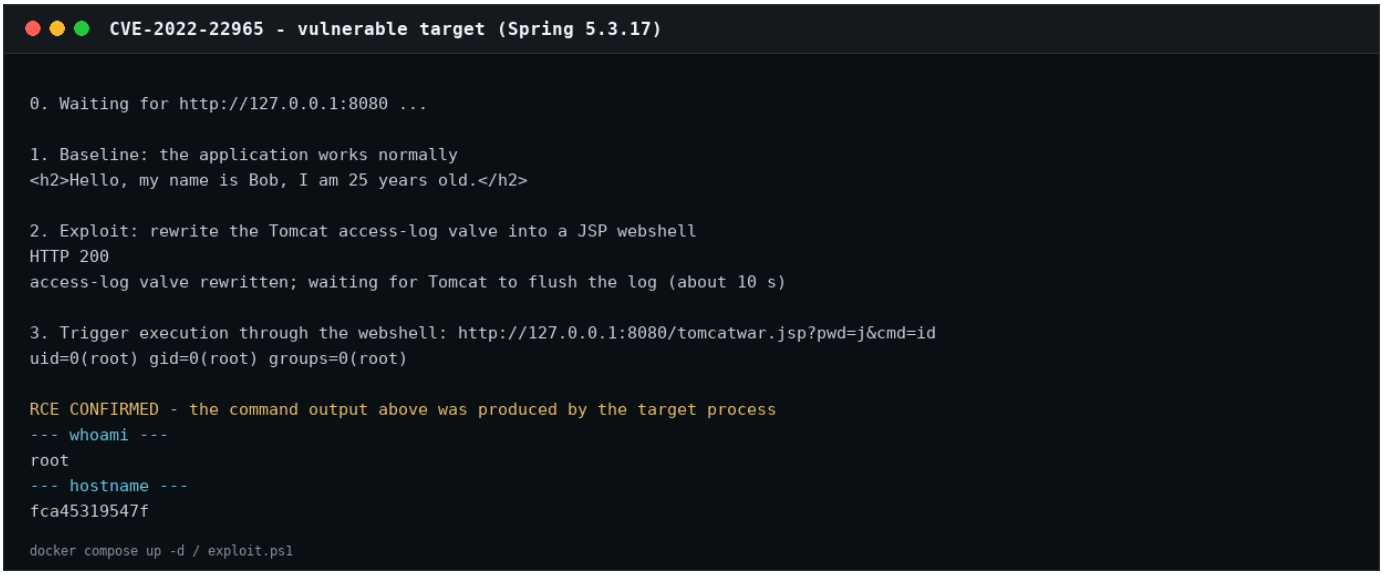


Figure 6 — the vulnerable target: baseline, the malicious binding request, and command execution through the generated JSP.

The request is a single `GET` to `/` with parameters whose *names* are the exploit:

```
class.module.classLoader.resources.context.parent.pipeline.first.pattern=<JSP>
class.module.classLoader.resources.context.parent.pipeline.first.suffix=.jsp
class.module.classLoader.resources.context.parent.pipeline.first.directory=webapps/ROOT
class.module.classLoader.resources.context.parent.pipeline.first.prefix=tomcatwar
class.module.classLoader.resources.context.parent.pipeline.first.fileDateFormat=
```

plus three headers (`c1: Runtime`, `c2: <%`, `suffix: %>//`) that the log pattern expands into JSP source at write time. The path walks the container's object graph:

```
class → module → classLoader → resources → context → parent → pipeline → first
  Class    Module   WebappCL    WebRoot     Context    Host      Pipeline  AccessLogValve
```

`first` is Tomcat's `AccessLogValve`. Rewriting `pattern`, `directory`, `prefix` and `suffix` makes the next access-log entry a JSP file inside `webapps/ROOT`; the log content is the servlet source, so requesting the file executes it. The empty `fileDateFormat` removes the date stamp from the filename, which is why the shell is reachable as `/tomcatwar.jsp`. Tomcat buffers logged lines (128 KiB), which is why the script waits ~10 s before triggering the file.

4.4 Root cause

Spring MVC binds request parameters to a model attribute:

```
@RequestMapping("/")
public String hello(Person person, Model model) { ... }
```

Before 5.3.18, `CachedIntrospectionResults` exposed the JavaBeans `class` property to that binder. From `Class` the attacker reaches the JDK 9+ `Module`, its `classLoader`, Tomcat's web resources, the `StandardContext`, its parent `StandardHost`, the host pipeline and the first valve. Every step is an ordinary public getter; nothing checks whether the *request* should be allowed to touch container internals.

The defect is therefore a **capability the binder should never have had**, not a missing authorisation check in the demo application. Two preconditions narrow it: the application must bind a POJO (not `@RequestParam` individual fields), and it must run on Tomcat as a WAR, because `resources.context.parent` is the servlet container's object graph. On JDK 8 the chain is shorter and this exploit does not work; a Spring Boot executable jar usually has no configured access-log valve, so the published sink is absent (the underlying primitive is still considered general by the advisory).

Impact: arbitrary command execution as the Tomcat process user. In this container that user is `root`, hence `uid=0(root)`; in a normal deployment it is the service account, which still owns the application, its database credentials and its files.

4.5 The fix

`cve/Dockerfile.fixed` rebuilds the same application with the Spring jars upgraded to **5.3.18**, the release that makes `class.*` (and `Class.*`, `*.class.*`, `*.Class.*`) non-bindable. The base image has no `jar` tool, so the Dockerfile unpacks `ROOT.war` into `webapps/ROOT` and replaces `WEB-INF/lib/spring-*.jar`; Tomcat deploys the exploded directory.

```
docker compose down
docker compose -f docker-compose.fixed.yml up -d --build
docker exec spring4shell-fixed unzip -p \
  /usr/local/tomcat/webapps/ROOT/WEB-INF/lib/spring-webmvc-5.3.18.jar \
  META-INF/MANIFEST.MF | head -5
```

Why this is the right fix: the change is in `CachedIntrospectionResults`, so the `class` property is no longer reachable *at all*. That removes the primitive, instead of blocking the one published payload. The advisory's workaround for applications that cannot upgrade — an `@InitBinder` that calls

`setDisallowedFields("class.*", "Class.*", "*.class.*", "*.Class.*")` — has to be repeated in every controller that binds input, which is why the framework upgrade is preferable.

What is **not** a fix: deleting the webshell (recreated on the next request), disabling the access log (loses auditing, keeps the primitive), blocking the literal string `class.module` in a WAF (trivial to vary), or running Tomcat as a non-root user (good hardening, but the webshell still executes with the application's privileges).

4.6 Verifying the fix

```
powershell -ExecutionPolicy Bypass -File .\verify-fix.ps1
```

```
1. Regression: the application still serves normal requests
GET /?name=Bob&age=25 -> HTTP 200
[PASS] normal request returns 200 and renders Bob, 25

2. Replay the exploit (Spring 5.3.18 ignores the class.* properties)
exploit request -> HTTP 200 (200 here means 'ignored', not 'exploited')

3. No webshell was written
GET /tomcatwar.jsp -> HTTP 404
[PASS] the JSP webshell does not exist (404)
[PASS] no *.jsp file in the exploded webroot

4. The access-log valve was not rewritten
last access-log line: 172.18.0.1 - - [...] "GET /?name=Regression&age=30 HTTP/1.1" 200 92
[PASS] the access log keeps the default pattern (request line, status, bytes)
[PASS] the log line does not contain JSP source

5. Binding of normal model properties still works
[PASS] argument binding for the normal model still works

RESULT: 6 passed, 0 failed
fix verified - all checks passed
```

CVE-2022-22965 - patched target (Spring 5.3.18)

```
1. Regression: the application still serves normal requests
GET /?name=Bob&age=25 -> HTTP 200
[PASS] normal request returns 200 and renders Bob, 25

2. Replay the exploit (Spring 5.3.18 ignores the class.* properties)
exploit request -> HTTP 200 (200 here means 'ignored', not 'exploited')

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GET /tomcatwar.jsp -> HTTP 404
[PASS] the JSP webshell does not exist (404)
[PASS] no *.jsp file in the exploded webroot

4. The access-log valve was not rewritten
last access-log line: 172.18.0.1 - - [20/Oct/2026:03:35:11 +0000] "GET /?name=Regression&age=30 HTTP/1.1" 200 92
[PASS] the access log keeps the default pattern (request line, status, bytes)
[PASS] the log line does not contain JSP source

5. Binding of normal model properties still works
[PASS] argument binding for the normal model still works

RESULT: 6 passed, 0 failed
fix verified - all checks passed

docker compose -f docker-compose.fixed.yml up -d / verify-fix.ps1
```

Figure 7 — the patched target: every check passes, including the two that a naive test would miss.

Why the verification does not check the status code. Spring 5.3.18 does not answer the malicious request with `400`; it silently ignores the non-bindable properties and returns the normal page (`200`). A test that asserts "the exploit must return 400" would therefore fail on a correctly patched server and would tempt someone to "fix" the test instead of the server. The script checks the *effects* instead:

1. no `tomcatwar.jsp` and no `.jsp` appears in the webroot;
2. the access log still has its default pattern, checked from inside the container — i.e. the valve was not rewritten;
3. the normal `name`/`age` binding still works, which catches a "fix" that simply broke the binder.

4.7 Defence in depth

- Keep Spring and Tomcat patched; treat framework CVEs as dependency work, which is what the SCA job in Part 1 automates.
- Bind to DTOs with explicit fields and use `@InitBinder` allow-lists (default-deny), not `setAllowedFields` on a few fields.
- Run the container as an unprivileged user with a read-only webroot.
- Put the application behind a WAF **and** keep it patched — the WAF buys time, it is not the fix.
- Alert on `class.module.classLoader` in request logs: legitimate clients do not send it.

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5. Conclusion

The two halves of the work meet in the same lesson. Part 1 shows that the protections the OWASP Top 10 asks for are implementable in a small service, and that each one is testable: a payload either authenticates or it does not, a script tag either survives to the client or it does not. Part 2 shows the other side of the same coin: a single framework feature can undo an entire application's security posture, and the only durable fix is to remove the capability, not the payload.

The most useful concrete finding is the verification trap. After the upgrade, the attack request still returns `200`; only the *effects* changed. Verifying a fix by status code would have declared a patched server vulnerable, and — worse — a superficial "fix" (deleting the JSP, disabling the log) would have passed. Checking the object graph's behaviour, and re-testing normal functionality, distinguishes a fix from a workaround.

Mapping to the grading rubric

Criterion	Points	Where
API functionality	2	§3.2, Figures 1–2: login, protected feed, create/update/delete, refresh
SQLi protection	0.5	§3.3; `tests/notes.test.js`
XSS protection	0.5	§3.4; escaping tests
Authentication (JWT + hashing)	1	§3.5; `tests/auth.test.js`, `tests/refresh.test.js`
CI/CD with SAST/SCA	2	§3.8, Figures 3–5; `.github/workflows/ci.yml`
Vulnerable environment + reproduction	2	§4.2–4.3, Figure 6; `cve/`
Root-cause depth	1	§4.4
Fix + verification	1	§4.5–4.6, Figure 7; `cve/Dockerfile.fixed`

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6. Control questions

1. Why is bcrypt (or scrypt/Argon2) preferable to SHA-256 for passwords? SHA-256 is designed to be fast: a GPU can compute billions of candidates per second, so a stolen digest is crackable at scale. bcrypt/scrypt/Argon2 are deliberately slow and salted: the salt defeats precomputed rainbow tables, the cost factor makes each guess expensive, and scrypt/Argon2 also require memory, which hurts GPU and ASIC attackers. This project uses scrypt with $N=2^{15}$, r=8, p=1 — about 80 ms and 32 MiB per hash, invisible for one login, ruinous for an offline attack.

2. SAST vs DAST? SAST analyses source or bytecode without running the program: it sees every branch, runs in CI before deployment, and reports the file and line — but it cannot see runtime configuration and produces false positives. DAST attacks a running application from outside: it finds real, reachable issues (including configuration and authentication problems) but only on the paths it crawls, and it cannot say where in the code the defect is. They complement each other; this work uses SAST in CI, and the companion DAST work uses ZAP.

3. How does a JWT work? A JWT is `base64url(header).base64url(payload).base64url(signature)`. The header names the algorithm, the payload carries claims (`sub`, `iss`, `aud`, `exp`, `iat`, `jti`), and the signature is an HMAC (HS256) or a public-key signature (RS256) over the first two parts. The server verifies the signature with its secret (or public key), then checks the claims — issuer, audience, expiry. The token is *signed, not encrypted*: anyone can read the payload, so it must not contain secrets. In this project the server also re-reads the user on every request, so deleting an account invalidates its token immediately.

4. What risks come from not auditing third-party dependencies? A project inherits every vulnerability of its dependency tree, including transitive packages nobody reviewed. Known CVEs are public and tooling makes them cheap to exploit (Spring4Shell is a single request). Without SCA you learn about the problem from an incident, not from CI. Auditing also exposes abandoned packages, licence conflicts and supply-chain risk (typosquatting, compromised maintainer accounts). `npm audit` in CI and OSV locally are the minimum.

5. What do Docker and Vulhub add for studying security? Isolation (a deliberately vulnerable service runs on a private network, not on the host), reproducibility (image + compose file pin the exact versions), speed (a container is destroyed and recreated in seconds, so an RCE can be repeated and the fix tested immediately), and safety (no vulnerable software installed on the workstation). Vulhub adds a curated, documented set of environments with the vulnerable versions and exploitation notes, which removes the setup work and leaves the analysis.

6. Vulnerability vs exploit — what was reproduced here? The vulnerability is the weakness (the binder exposing `class.module…`); the exploit is the concrete sequence that abuses it (the parameter names, the headers that build the JSP, the request to the generated file). In this work the *vulnerability* is CVE-2022-22965 in Spring 5.3.17; the *exploit* is the HTTP request and webshell produced by `exploit.ps1`. Reproducing the exploit proves the vulnerability is reachable in this configuration.

7. Why is regression testing critical after a fix? A patch changes behaviour by definition, and security patches often touch shared code. Without a regression check you cannot distinguish "the attack is blocked" from "the feature is broken". Here the fix could easily have been implemented by disabling model binding or the access log, which would block the exploit and break the application. `verify-fix.ps1` therefore also asserts that `/?name=Bob&age=25` still renders correctly.

8. Why does CVE knowledge matter for a pentester or a SOC analyst? For a pentester, a CVE gives a known-reachable primitive with public proof and a precise version range — the fastest path from "this looks like Spring" to "this is exploitable". For a SOC analyst, the same knowledge drives detection: the request signature (`class.module.classLoader`), the post-exploitation artefact (a new `.jsp` in the webroot) and the upgrade timeline. Both roles need to reproduce and verify, otherwise they are repeating rumours.

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7. Submission checklist

Item	Where
Public repository with the project	`https://github.com/faris-faris-faris/work1-secure-notes-api`
README with API, security measures, CVE, root cause, fix, proof	`README.md`
Link to the last successful pipeline run	`https://github.com/faris-faris-faris/work1-secure-notes-api/actions/runs/40103379944`
SAST/SCA screenshots	Figures 3–5, `docs/screenshots/`
CVE reproduction with commands and screenshots	§4.2–4.3, Figure 6
Root-cause analysis	§4.4
Fix and proof of elimination	§4.5–4.6, Figure 7, `cve/Dockerfile.fixed`
Folder with the fixes	`cve/`
Reports (EN + ZH, PDF)	`docs/report_en.pdf`, `docs/report_zh.pdf`
Evidence transcripts	`docs/evidence/`

> The title page carries the student's name (Wu ZiYang) and HDU number (23320121); no variant was assigned ("—"). The repository is public and the pipeline run above is the last successful one.